

What's driving open source

An open source project is released publicly, with design files available online for download with a defined sharing licence which allows designers to replicate, hack, improve, modify and contribute to the project. Of course, there are some factors driving hobbyists, electronic enthusiasts and even professional designers to replicate, improve and build stuff. Let's find out what's motivating them to do so.

Cost-effective tools enabling open system designs. With respect to India, although open source electronics is still at a nascent stage, Kavita Arora, founder, Bangalore MakeSpace and Open Source Creativity, says, "Electronic components and embedded computing modules are getting more affordable today, which is definitely one of the most interesting changes in this arena." This cost-effectiveness is motivating hobbyists to design and build systems in a budget-friendly manner. Popular boards like Arduino are also copied and built into non-branded (such as Freeduino) boards that sell at a lower cost at sites like kitsnspares.com

Sharing his personal experience, Sivaram notes, "When I used to buy boards, an Arduino, for instance, cost me around 1100 rupees. Keeping aside the Chinese clones of Arduino, Indian boards are available today for 600 to 700 rupees. The hobbyists I meet today are buying these pocket-friendly, India made boards which enable them to build many projects." This is really driving the costs down and, in turn, helping a lot of people build stuff using open source hardware.

Communities collaborate and innovate. More than the availability of cheap hardware what motivates people is actually the people around them, believes Kuber. So, more of competition, clubs and communities are driving people to build stuff using open source electronics. She says, "We have an open house every Friday at the Arduino office which is open to everyone. We teach enthusiasts on various

Highlights of Electronics Rocks 2014 (Open Design track)

October 10-11, 2014 at NIMHANS Convention Center, Bengaluru

1. **Pixhawk Fire: Open source autopilot for drones on Linux.** Starting with a test run to check out some manoeuvres, go over to its development process, applications and openness.

2. **Open source for automation and robotics in industrial sector.** This session will highlight basics of the open architecture for industrial/consumer robotics and automation, and some demonstrations explaining the working nature of the protocols and RTOS for robotics. Will include real-time implementation concepts on automation and knowledge about drivers, along with the proof of concepts/practical demos.

3. **Using e-scrap and Android AppInventor for rapid prototyping.** For those who, although unfamiliar with core electronics and programming, wish to design industry-level products. Will cover parts of e-scrap you can use, such as non-working cell phones, cell-phone chargers and motherboards. Application design tools will also be explained.

4. **Coolest applications on the coolest boards.** Comparison of popular development boards like Arduino, Raspberry Pi, RIoT, Galileo, Goosberry and UDOO, and for which kind of applications each one is suitable.

5. **Workshop: Let's Get Started With 3D Printing.** A workshop on getting started with 3D printers, the software needed, creating mechanical designs and more hands-on elements for newbies to printing.

6. 3D printing will change how the world learns

- Rapid prototyping and its benefits
- Operating 3D scanners, 3D printers and 3D printing web portals
- Developing innovative 3D printed products
- Supporting defence sector with advanced technology
- Increasing employment, killing unemployment
- Reducing cost of goods

7. **Low-cost rapid prototyping for Indian situations.** Aakash, the founder of Aha 3D, will share his experiences while working on sophisticated 3D printing technology. Useful for those who want to build viable business out of low-cost/DIY 3D printing.

8. **From paper to life: Additive and subtractive rapid prototyping.** Besides introducing the basic concepts of rapid prototyping, it will introduce you to the applications of 3D printers and CNCs as a part of rapid prototyping.

Other factors driving the open source hardware community

1. Platforms like Arduino, Raspberry Pi, Plugcomputer, BeagleBoard and LaunchPad have become de-facto standards. All other semiconductor companies are coming up with add-on/daughter cards that are compatible to either of these platforms.

2. There are disruptive changes happening, especially in design space, due to open source electronics. Prototyping hardware cost has been reduced by nearly 50% due to use of open source electronics versus evaluation/development kits (such as Anjan Contractor built 3D Food Printer using RepRap.org).

3. Accessibility to different kinds of 'silicon' from a variety of semiconductor companies has been a major benefit. Open source software community has become more active than ever in order to provide latest kernels, drivers, libraries and applications that enable good support for such open source electronics. Major beneficiaries are universities and students who are able to use these devices to make very interesting projects.

4. TI has taken additional measures by providing a reference design library. TI Designs provides all design files for reference designs, including bill of material, enabling designers take benefit of this and go to market quickly.

Source: Dhaval Vasa, Solution Architect, elfinshops

methods to interface one technology to an Arduino. One week we have session on Bluetooth, another week we have on GSM and the next we focus on touch interface, and so on." Kuber adds, "If people come together and learn, it boosts the engagement and the learning process."

There is a lot of buzz among people to play with open source hardware and create something. A lot of them are attending sessions actively

organised by various communities, enabling them to engage and learn efficiently. Sivaram adds, "Last time the IoTBLR group was having a demo session, there were around 150 people put up in a hall. I also administer a Facebook group called Open Source Hardware Bangalore (OSHWB). It has 700-800 members. So there is definitely a lot of interest."

Talking about Electronics Rocks (eRocks) held in Bengaluru, Varun